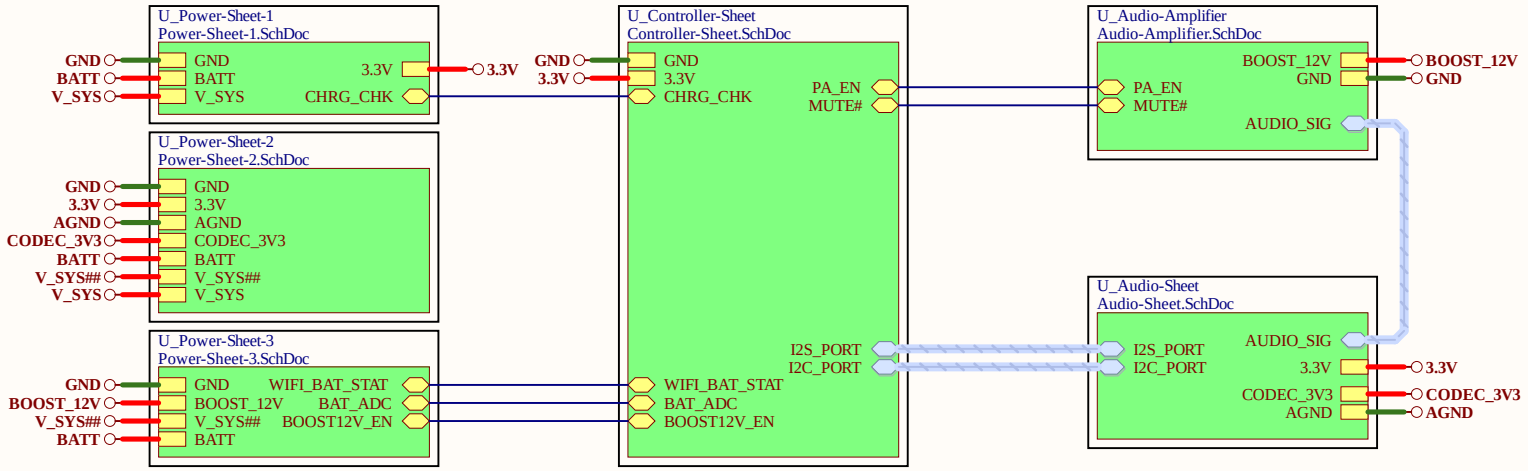
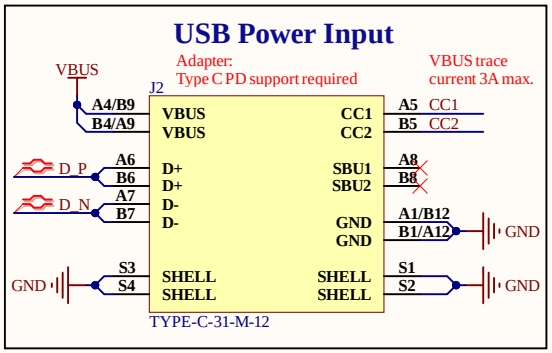
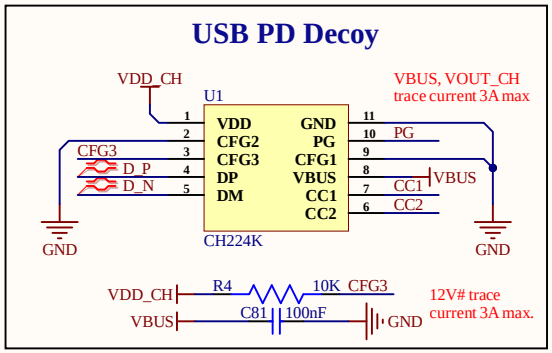
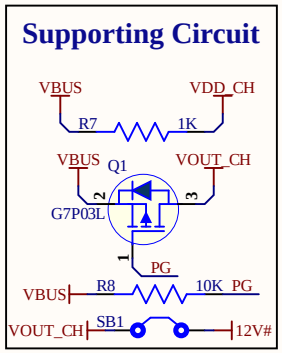


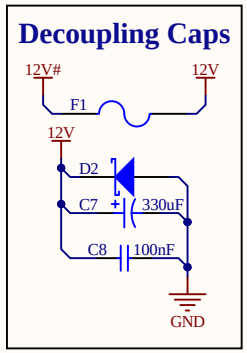
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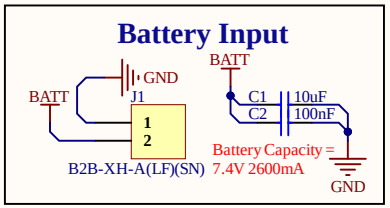
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Date: 3/20/2023	Page Num: 1
Drawn By: Huzaifa Ali	Page Total: 7

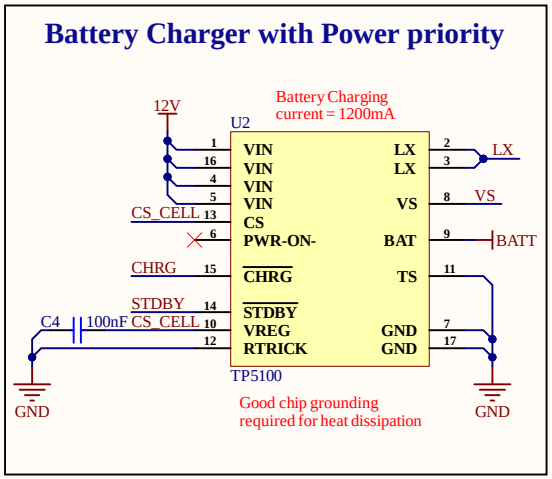


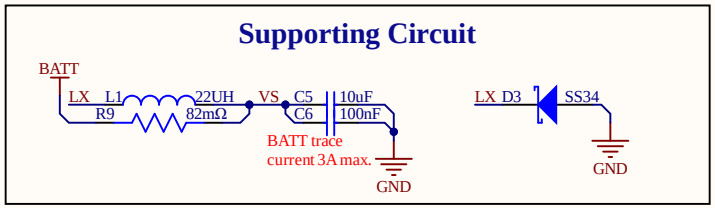


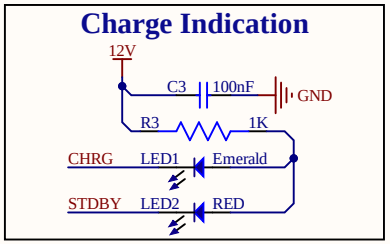


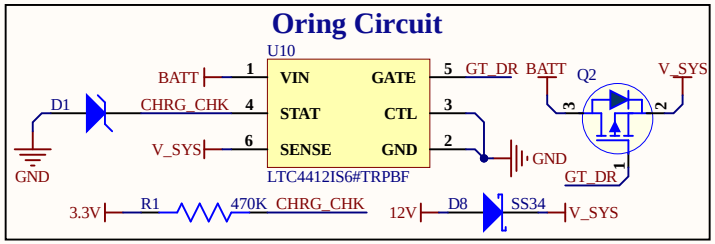


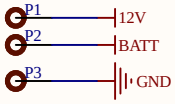


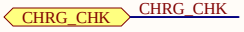


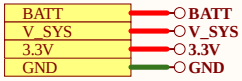












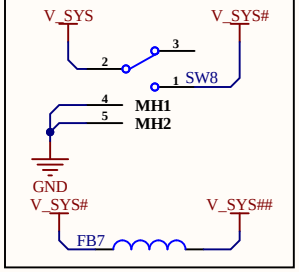
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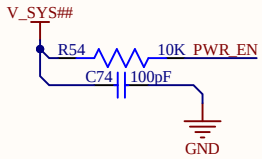
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Date: 3/20/2023	Page Num: 2
Drawn By: Huzaifa Ali	Page Total: 7

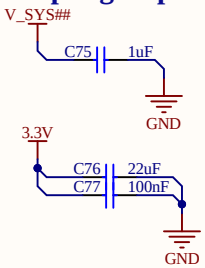
Power Switch



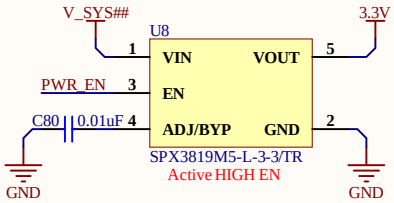
Supporting Circuit



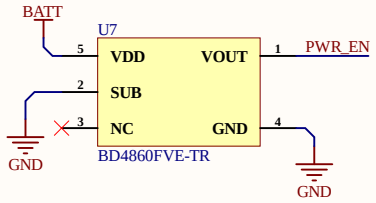
Decoupling Capacitor



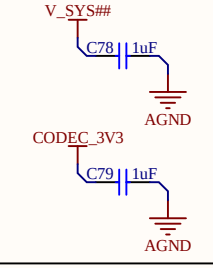
BAT to 3.3 LDO



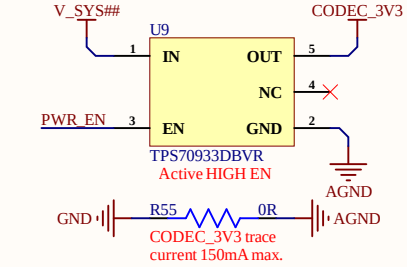
LOW Voltage Detector



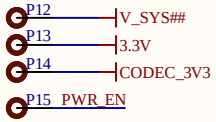
Decoupling Capacitor

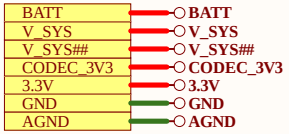


BAT to 3.3 Low Noise LDO



Test Points





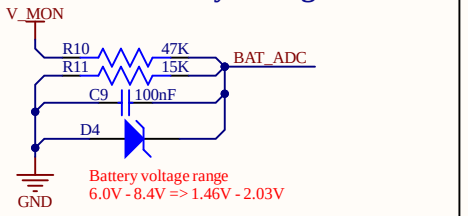
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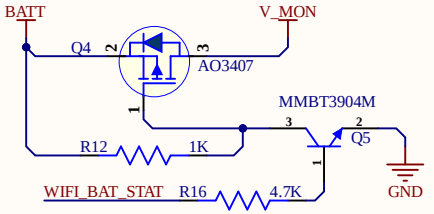
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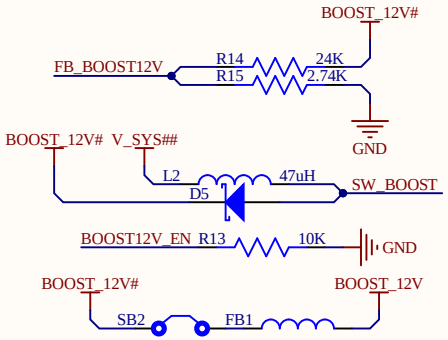
Date: 3/20/2023	Page Num: 3
Drawn By: Huzaifa Ali	Page Total: 7

Read Battery Voltage

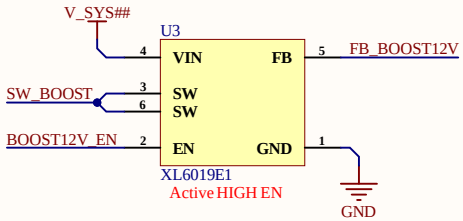




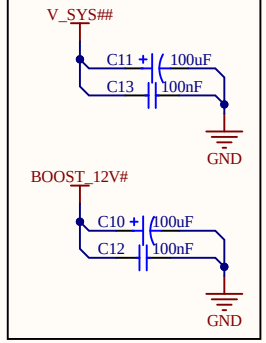
Supporting Circuit



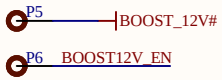
Boost Converter

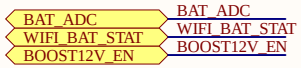


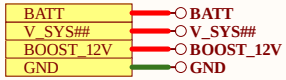
Decoupling Caps



Test Points



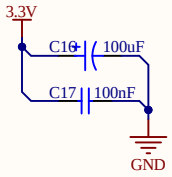




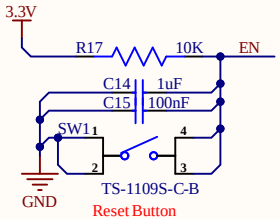
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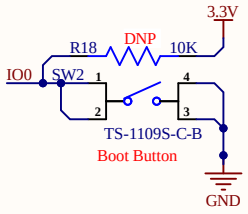
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Decoupling Caps

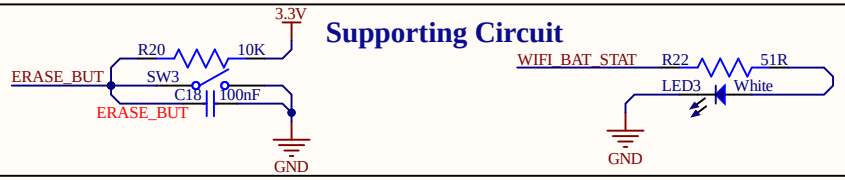


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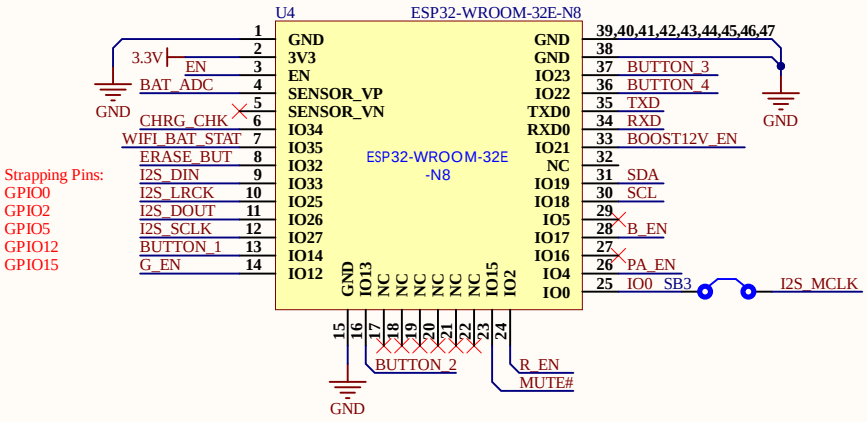




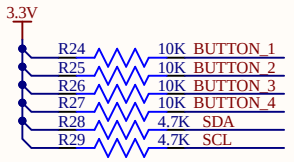
Supporting Circuit



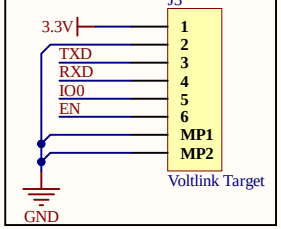
ESP32 Controller



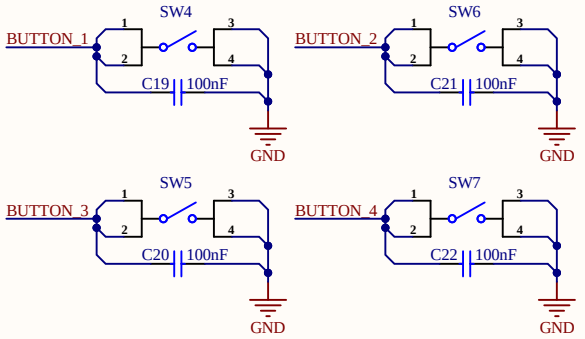
Supporting Pull-ups



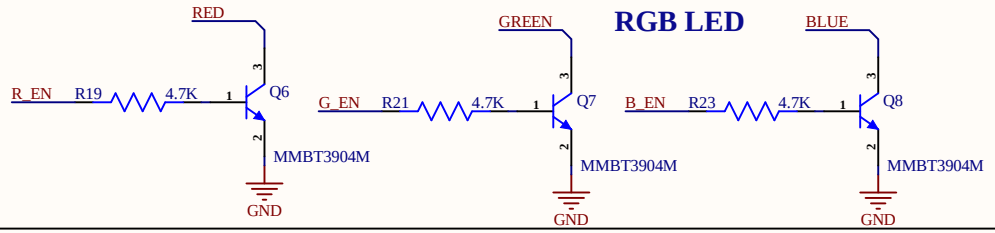
Programming

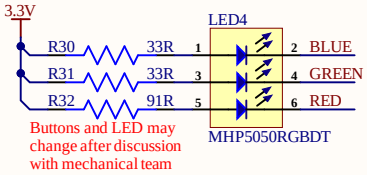


Button Inputs



RGB LED



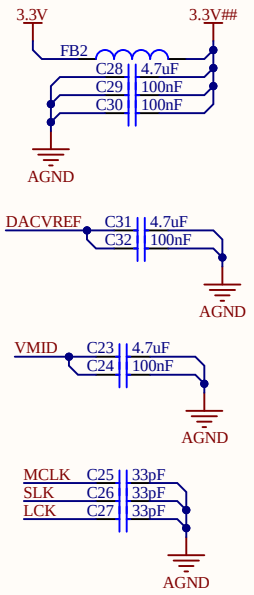




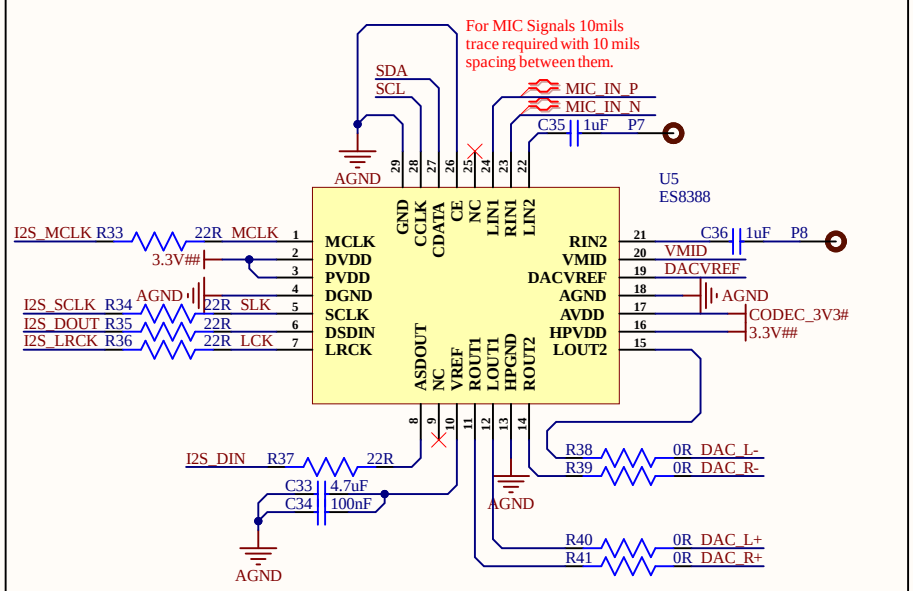
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Date: 3/20/2023	Page Num: 5
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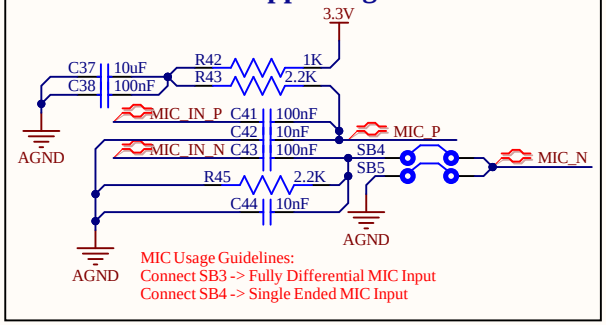
Decoupling Caps



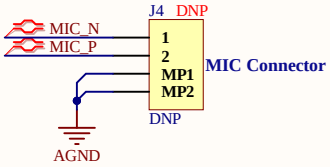
Audio Codec



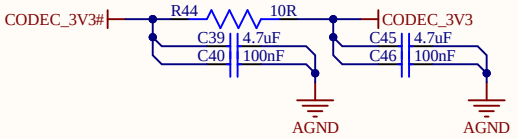
MIC Supporting Circuit

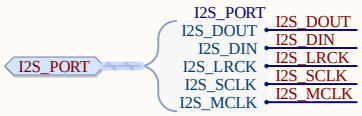


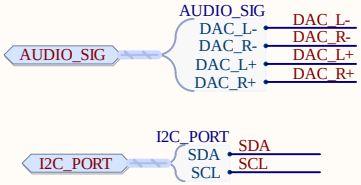
MIC Connector

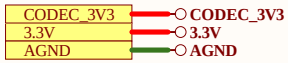


Supporting Circuit





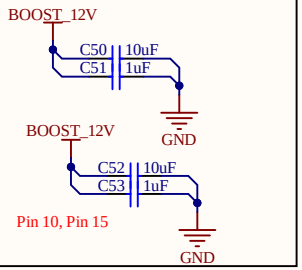




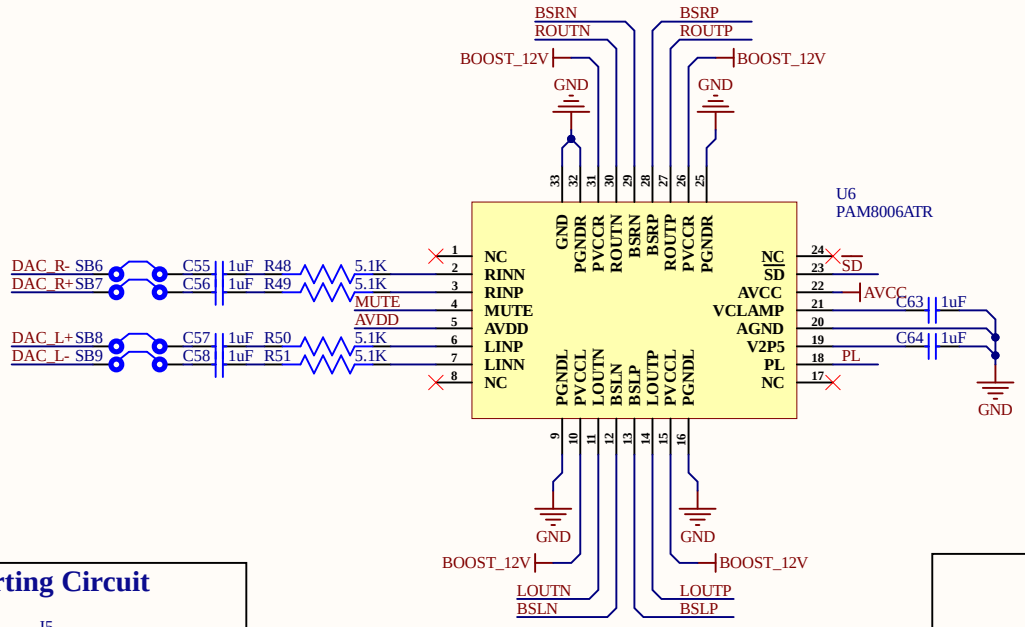
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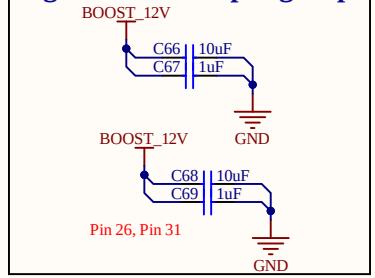
Left Side Decoupling Caps



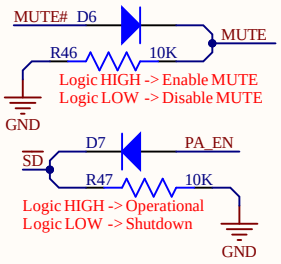
Audio Amplifier



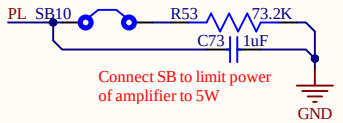
Right Side Decoupling Caps



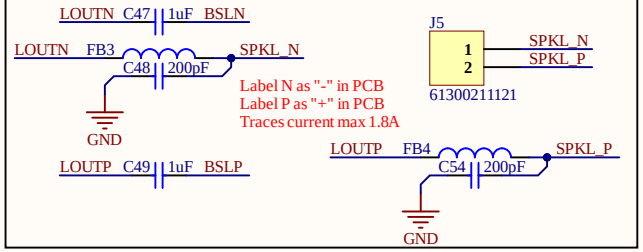
Supporting Circuit



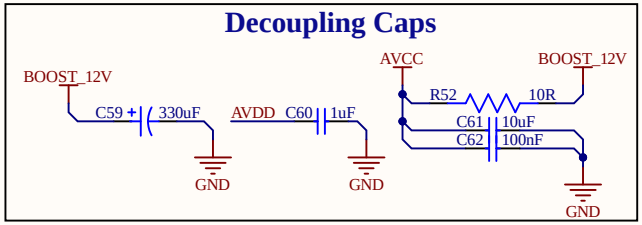
Power Limit



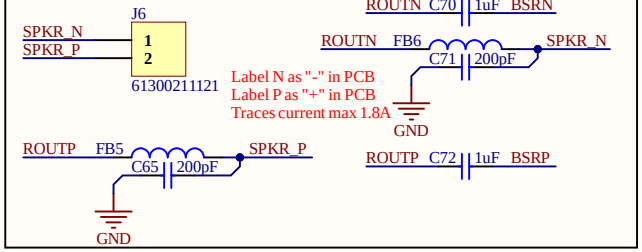
Left Speaker Supporting Circuit



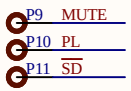
Decoupling Caps



Right Speaker Supporting Circuit



Test Points



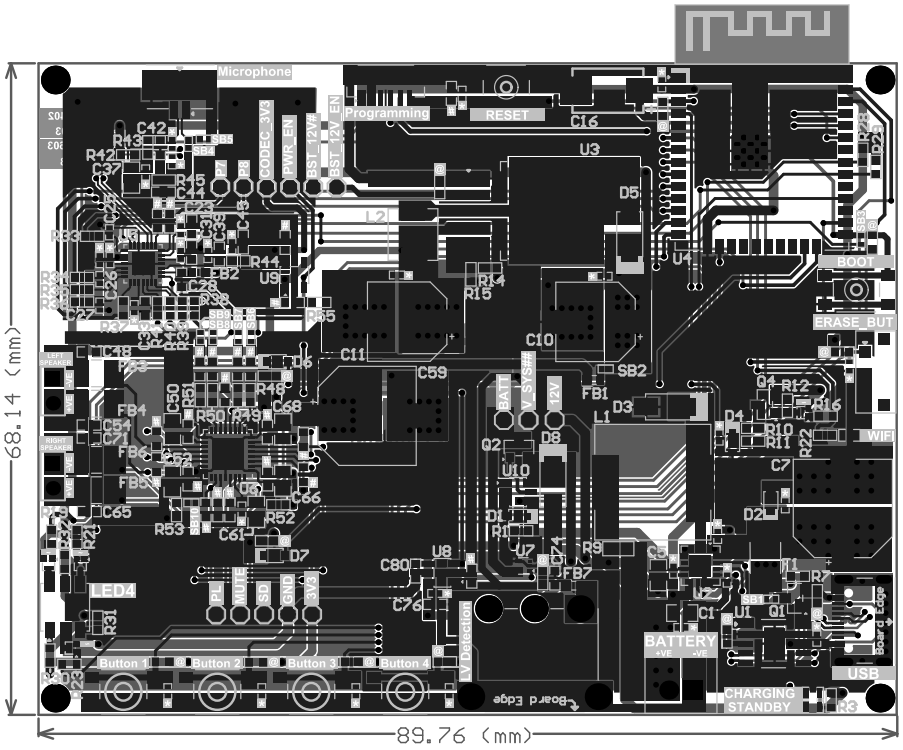






Cannot open file
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Title: Audio-Amplifier.SchDoc	
File: IoT-Speaker.PrjPcb	
Date: 3/20/2023	Page Num: 7
Drawn By: Huzaifa Ali	Page Total: 7



Layer	Name	Material	Thickness	Constant	Board Layer Stack
	Top Overlay				
	Top Solder	Solder Resist	3.00mil	3.5	
1	Top Layer		10.00mil		
	Dielectric 1	FR-4	37.00mil	4.8	
2	Bottom Layer		10.00mil		
	Bottom Solder	Solder Resist	3.00mil	3.5	
	Bottom Overlay				

Comment	Description	Designator	Footprint	LibRef	Quantity
10uF	CAP C1R 10UF 25V X5R 0805	C1, C9, C37, C50, C52, C61, C66, C68	Cap-0805	CC0805KKX5R6BB106	8
100pF	CAP C0G 0.1UF 25V X5R 0402	C2, C3, C4, C6, C8, C9, C12, C13, C15, C17, C18, C19, C20, C21, C31, C34, C39, C38, C32, C36, C35, C46, C41, C43, C45, C62, C77, C85	Cap-0402	CC0402KRX5R6BB104	29
100uF	CAP ALUM POLY 100UF 20% 35V SMD	C7, C59	350xRY330M	350xRY330M	2
100uF	CAP ALUM POLY 100UF 20% 35V SMD	C10, C11	350xRY100M	350xRY100M	2
1uF	CAP C0G 0.1UF 25V X5R 0603	C14, C16, C26, C47, C49, C53, C55, C56, C57, C58, C60, C63, C64, C67, C69, C70, C72, C73, C74, C79	Cap-0603	CC0603KRX5R6BB105	23
220uF	CAP TANT 330UF 20% 25V 2917	C16	Case-E	TAJE337M035RNJ	1
4.7uF	CAP C0F 4.7UF 10V 0805	C23, C30, C31, C33, C48, C78	Cap-0603	CC0603KRX7R7BB475	6
20pF	CAP C0G 20PF 25V 0402	C25, C81, C27	Cap-0402	CRA0102C1C200JRN3	3
10nF	CAP C0G 10000PF 50V 0603	C42, C44	Cap-0603	C1608X5RHP0C6MHA103	2
330pF	CAP C0G 330PF 50V 0603	C45, C54, C65, C71	Cap-0603	C1608X5R1H7R6MHA201	4
100pF	CAP C0F 100PF 50V 0603	C74	Cap-0603	C1608X5RHP0C6MHA101	1
22uF	CAP C0F 22UF 10V 1206	C76	Cap-0603	C1608X5R1H106MHA103	1
0.01uF	CAP C0F 10000PF 50V 0603	C80	Cap-0603	C1608X5R1H103KHA01	1
MMBZ5V3B	DIODE ZENER 3.3V 350MW SOT23-3	D1, D4	SOD323F	MMBZ3V3B	2
RSD12V/S1UA_115	TVS DIODE 12VWM 17VC SOD323	D2	SOD-323	RSD12V/S1UA_115	1
S1M	50mA@400V 400 Single 1A 0.52V@1A 60ns(SOD-214AC) Schottky Barrier	D3, D5, D6	SMA	S1M	3
BAS140WS 7-F	DIODE SCHOTTKY 40V 200MA SOD323	D6, D7	SOD-323	BAS140WS-7-F	2
SDMD1616J2-390-13.2V	13.2V 2.6A 40A 600mW -40°C ~ +85°C 3.3A 10mΩ 2.5x 32mΩ 1612 Resettable Fuses-SCHG	F1	1812-FUSE	SDMD1616J2-390-13.2V	1
SLMU60G-121TN1D	0.05OHTS 60A/D 120 Ω@4.00GS 0.LN	F201, F812, F87	1AD-0603 (1608 Metric)	SLMU60G-121TN1D	3
1Ohm-0225-4-MD1210	0.05OHTS 60A/D 600 Ω@4.12GS 0.LN	F83, F84, F85, F86	1AD-1210	1Ohm-0225-4-MD1210	4
3638-XX-A(LF)(SN)	3.2mm Shrouded Header (4 Sides) H2R 2 POS 2.54mm Solder ST Thru-hole Box	J1	3638-XX-A(LF)(SN)_	3638-XX-A(LF)(SN)	1
TYPE-C-31-M-12	2GbHA 1 1/0 Female Type-C SMD USB Connectors-SCHG	J2	TYPE-C-31-M-12_	TYPE-C-31-M-12	1
KeyStone Target	1x4P SH-1.0mm-4 SMD,P=1mm Wire To Board / Wire To Wire Connector-SCHG	J3	KeyStone Target	5145863	1
2NKP	1x2P SH-1.25mm Wire to 2 -40°C ~ +85°C Surface Mount SMD,P=1.25mm Wire To Board / Wire To Wire Connector-SCHG	J4	1.25-2PWB	1.25-2PWB	1
61300311121	CONN HEADER VERT 4POS 2.54MM	J5, J6	40HV1X_	61300311121	2
2R2H	2.2R 20uH ±20% 40mΩ SMD,12x12x6mm Power Inductors-SCHG	L1	PPIH1204-2R2M_	PPIH1204-2R2M	1
47uH	5A 47uH ±20% 57.5mΩ SMD,13.8x12.6x6.6m m Power Inductors-SCHG	L2	FSPI1365-470M	FSPI1365-470M	1
Emerald	Emerald 0603 Light Emitting Diodes (LED)-SCHG	LED1	LED-0603	NCD0603G2	1
GG3	0603 Light Emitting Diodes (LED)-SCHG	LED2	LED-0603	NCD0603R3	1
White	White figure 0603 Light Emitting Diodes (LED)-SCHG	LED3	LED-0603	NCD0603W3	1
MHPS0606RGB3T	2mcd White Diffused -40°C ~ +85°C RGB 6x6x3.55(H) Light Emitting Diodes (LED)-SCHG	LED4	SMD5050	MHPS0606RGB3T	1
TEST POINT	Test Point ON	P1, P2, P3, P5, P6, P7, P8, P9, P10, P11, P12, P13, P14, P25	Test-points	TEST POINT	14
STP03L	40V 7A 28mΩ 1.5W 1.1V@250uA P-Channel SOT-23-3L MOSFETs-SCHG	Q1	SOT-23-3L_	STP03L	1
AO3407	30V 4.1A 60mΩ@10V, 4.1A 102mW 2V@250uA P-Channel SOT-23 MOSFETs-SCHG	Q3, Q4	SOT-23	AO3407	2
MHMBT3904M	40V 100mW 1mA@10mA,1V 200mA 300MHz 900mV@50mA,5mA 6pF 67°C ~ +150°C (BVCE) SOT-723 Bipolar Transistors - BJT-SCHG	Q5, Q6, Q7, Q8	SOT-723	MHMBT3904M	4
470K	RES SMD 470K OHM 5% 1/10W 0603	R1	RES-0603	RC0603J0-470KRL	1
1K	RES SMD 1K OHM 5% 1/10W 0603	R3, R7, R12, R42	Res-0603	RC0603F0-0710KL	4
10K	RES SMD 10K OHM 1% 1/10W 0603	R4, R8, R13, R17, R18, R20, R24, R25, R26, R27, R46, R47, R54	Res-0603	RC0603F0-0710KKL	13
100mΩ	100mW Thick Film Resistors 150V ±5% ±400ppm/°C -55°C ~ +155°C 40mΩ 0805 Chip Resistor - Surface Mount-SCHG	R9	RES-0805	0805WA0100JLTSL	1
47K	RES SMD 47K OHM 0.5% 1/10W 0603	R10	Res-0603	RT0603BRD0747KL	1
15K	RES 15K OHM 1% 1/10W 0603	R11	Res-0603	RC0603F0-0715KL	1
24K	RES 24K OHM 1% 1/10W 0603	R14	Res-0603	RC0603F0-0724KL	1
2.74K	RES 2.74K OHM 1% 1/10W 0603	R15	Res-0603	RC0603F0-0727K7L	1
4.7K	RES SMD 4.7K OHM 1% 1/10W 0603	R16, R43, R21, R23, R28, R29	Res-0603	RC0603F-0747KL	6
1MR	RES SMD 1MR 5PPM 1/10W 0603	R22	Res-0603	RC0603F0-07E4RL	1
10R	RES SMD 0603 33 Ohm 1% 1/10W	R30, R31	Res-0603	RC0603F0-0733RL	2
1K5	RES 1K5 OHM 5% 1/10W 0603	R32	Res-0603	RC0603F0-07E5L	1
22R	RES 22 OHM 5% 1/10W 0603	R33, R34, R35, R36, R37	Res-0603	RC0603F0-0722RL	5
0R	RES 0 OHM JUMPER 1/10W 0603	R38, R39, R40, R41, R45	Res-0603	RC0603J0-0705L	5
2.2K	RES 2.2K OHM 1% 1/10W 0603	R43, R44	Res-0603	RC0603F0-072K2L	2
10R	RES SMD 10 OHM 1% 1/10W 0603	R44, R52	Res-0603	RC0603F0-0710RL	2
5.1R	RES 5.1R OHM 5% 1/10W 0603	R48, R49, R50, R51	Res-0603	RC0603F0-075K1L	4
73.2K	RES 73.2K OHM 1% 1/10W 0603	R53	Res-0603	RC0603F0-0773K2L	1
Solder Bridge 0402	Solder Bridge ON	SB1, SB2, SB3, SB4, SB5, SB6, SB7, SB8, SB9, SB10	Solder Bridge 0402	Solder Bridge 0402	10
TS-1109S-C-B	No NO 50mA 4.5mm 100000 260gf 12V -40°C ~+80°C 4.5mm 6.2mm Black Round Button Right Angle SPST SMD Tactile Switches-SCHG	SW1, SW2	TS-1109S-C-B	TS-1109S-C-B	2
ST-1113RA-P	-25°C ~+85°C 3MOS(locking Black Rectangle button 4mm Surface Mount, SPST 100MΩ 100000 260gf@vt@gf 12V SMD Tactile Switches-SCHG	SW3	ST-1113RA-P_	ST-1113RA-P	1
TI-1109S-B-J	No NO 50mA 4.5mm 100000 160gf 12V -40°C ~+80°C 4.5mm 6mm Black Round Button Brick-hugging SPST SMD Tactile Switches-SCHG	SW4, SW5, SW6, SW7	TI-1109S-B-J_	TI-1109S-B-J	4
SI-1201LS	Lying 1M SPDT 100V 0.3mm 1G 5mm 100V 0.5mm 3A PCPcs 60000Hectorings-M oilburnish -40°C ~+85°C Black Plugin Slide Switches-SCHG	SW8	SI-1201LS	SI-1201LS	1
24224K	24M 8L 4x2.2 L 3.0(SDP-SCHG	U1	8150DP-10	24224K	1
TP5100	TQFN-16-EP(4x4) Battery Management ICs-SCHG	U2	TQFN-16-EP(4x4)_	TP5100	1
AU8019E1	DC/ Nonsol converterType-1 board 65V 5V~40V 5A 100kHz TO-263-5 DC-DC Converters-SCHG	U3	TO-263-5 (D2PAK)	AU8019E1	1
SOT32-WROOM-32E-N8	SMD, 18x25.5mm WiFi Modules-SCHG	U4	38-SMD Module	SOT32-WROOM-32E-N8	1
816088	QFN-28-EP(4x4) MCU/DAC - Specialized-SCHG	U5	QFN-28-EP(4x4)	816088	1
PAM8006AATR	IC AMP AUDIO 15W CLASS D 32QFN	U6	32-QFN	PAM8006AATR	1
SD4880F-VC-TR	IC SUPERVISOR 2 CHANNEL SOT23	U7	VSOF-6	SD4880F-VC-TR	1
SPX3819M5-L-3-3/TR	IC REG LDO 3.3V 0.8A SOT23-5	U8	SOT-23-5	SPX3819M5-L-3-3/TR	1
PT5700S30MVR	DC/DC Step-Down 2.3V 100mA SOT23-5	U9	SOT-23-5	PT5700S30MVR	1
TC4413VOA713F6F	IC GR CTRLR 9MC SOJ/ET TSOT23-8	U10	SOT-23-6	TC4413VOA713F6F	1